Meta-Kaizen Series · Paper 5 of 5 · Capstone

On the Decision to Act: Strategic Convergence and the Mathematics of Intervention Timing at System Tipping Points

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AI contributions: Claude (Anthropic) provided mathematical formalization, historical synthesis across seven strategic traditions, literature review, and manuscript writing. Grok (xAI) provided the AMOC empirical data processing and quantitative validation for the pre-specified test case. All general theory and mathematical formulations are contributed to the public domain.

Series Orientation

This is Paper 5 of five in the Meta-Kaizen series, and the final paper. It is the capstone.

Paper 1 derived the axiomatic foundation of the Meta-Kaizen governance framework and the Kaizen Variation Score (KVS), with the approval threshold kappa = 0.50 derived under a symmetric loss assumption. Paper 2 proved network stability and governance closure across distributed implementations. Paper 3 formalized the reasoning and memory architecture that allows the framework to learn from its own history without being trapped by it. Paper 4 introduced the Fractal Rhythm Model (FRM) as an adaptive layer that detects regime shifts and modifies the KVS computation accordingly, introducing Axiom 5-prime as a formal departure from Paper 1's stationarity assumption.

Each of these papers deferred a question it could not answer. Paper 1 deferred: what if false positives and false negatives do not cost the same? Paper 2 deferred: why are distributed networks more resilient than centralized ones, beyond the mathematical proof of stability? Paper 3 deferred: what triggers the decision to adapt when a signal emerges? Paper 4 deferred: once a regime shift is detected, when should the agent act?

FRM v5 added climate tipping point detection as the highest-Jaccard bisociation domain in the series, bringing the Early Warning Signals (EWS) literature into contact with the Sentinel architecture. That contact revealed a documented gap: the EWS literature can detect approaching transitions with increasing reliability, but has no principled decision theory for when to act on those detections. A 2025 review in the Proceedings of the Royal Society B confirmed the gap formally: EWS systems can paradoxically induce riskier behavior when no signal is received, because agents become overconfident in the absence of warning.

This paper closes the gap. It derives a decision-theoretic framework for intervention timing at system tipping points. It shows that seven independent traditions of strategic thought, spanning 2,500 years and six civilizations, independently converged on the same five-part structure for acting effectively under transition uncertainty. It formalizes that structure in four theorems. And it

connects every deferred question from Papers 1 through 4 to a formal result. The series now has its ending.

Abstract

The Early Warning Signals (EWS) literature has achieved substantial progress in detecting the statistical precursors of critical transitions in complex adaptive systems. What it has not achieved is a principled decision theory for when to act on those detections. This paper addresses the gap.

We identify a convergence across seven independent traditions of strategic thought -- Sun Tzu (c. 500 BCE), Thucydides (c. 400 BCE), Machiavelli (1513), Clausewitz (1832), Liddell Hart (1929), Boyd (1976), and Dowding (1936-1940) -- on a five-part structure for effective action under transition uncertainty: signal reading, uncertainty quantification, the bounded intervention window, distributed network architecture, and decision threshold. These traditions were separated by culture, century, and domain; their convergence on identical structure is evidence that the underlying problem is universal.

We formalize this structure in four theorems. Theorem 1 (Window Rationality) proves that intervention is rational if and only if the expected detection window width exceeds a threshold determined by the agent's cost structure and the system's own transition-time uncertainty. Theorem 2 (Asymmetric Loss Threshold) derives the optimal Sentinel detection threshold under asymmetric loss, showing that Paper 1's kappa = 0.50 approval threshold is the symmetric special case of a general result. Theorem 3 (Distributed Detection Advantage) proves that k independent monitoring nodes extend the detection window by (1/lambda)*(1 - 1/k) relative to a single observer, saturating at 1/lambda in the large-k regime, with the historical Dowding system providing a quantitative consistency check. Theorem 4 (Self-Generated Friction) proves that as a system approaches its tipping point, the coefficient of variation of the tipping time diverges, creating a point t_trap before the tipping time beyond which intervention is no longer rational regardless of cost structure.

The paper connects each theorem to the FRM Sentinel architecture, resolves each deferred assumption from Papers 1 through 4, and provides a pre-specified empirical test case using AMOC (Atlantic Meridional Overturning Circulation) data. It constitutes the decision-theoretic capstone of the Meta-Kaizen series.

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1. Introduction

1.1 The Missing Half of Early Warning Science

The detection of critical transitions in complex adaptive systems has become one of the most active areas of research in ecology, climate science, neuroscience, and financial systems. Building on the theory of critical slowing down (CSD) near fold bifurcations, researchers have developed a battery of statistical indicators -- rising variance, increasing autocorrelation, diverging detrended fluctuation analysis exponents -- that reliably precede abrupt regime shifts in a wide range of systems (Scheffer et al. 2009; Dakos et al. 2012; Lenton et al. 2012; Boers 2021).

What the literature has not developed, with comparable rigor, is the decision theory that should govern action in response to these detections. The detection problem and the decision problem are not the same problem. Detecting that a tipping point is approaching tells you the system is vulnerable. It does not tell you when a rational agent should intervene, what detection sensitivity is appropriate given the costs of false alarms versus missed transitions, or whether waiting for a stronger signal is better or worse than acting on a weaker one.

The consequences of this gap are not merely theoretical. A 2025 review published in the Proceedings of the Royal Society B found that EWS-informed decision systems can paradoxically induce riskier behavior: agents who receive no warning signal become overconfident in the system's stability and increase their exposure precisely when the system may be approaching a regime boundary. The detection apparatus, without accompanying decision theory, can make outcomes worse than no detection at all.

This paper provides the missing decision theory. It derives four theorems that characterize when intervention is rational, what detection threshold is optimal under asymmetric loss, how distributed monitoring architecture extends the actionable window, and why the window closes inexorably as the tipping point approaches. The theorems are derived from first principles in sequential decision theory. They are then shown to be independently anticipated -- qualitatively, across six civilizations and 2,500 years -- by the great traditions of strategic thought.

1.2 The Strategic Traditions' Unformalized Answer

The proposition that strategic thinkers anticipated formal results in decision theory and complexity science is not new. Beyerchen (1992), in a landmark paper in International Security, demonstrated that Clausewitz's treatment of friction and nonlinearity in On War is structurally isomorphic to modern nonlinear dynamics. Friedman (2022), writing in the Naval War College Review, extended this analysis to show that Clausewitz's conceptual architecture anticipates complex adaptive systems theory more broadly.

This paper extends that line of argument to a claim that is both broader and more specific. Broader because it encompasses seven traditions rather than one. More specific because it identifies a precise five-part structure -- not a loose family of similarities -- that all seven traditions independently found. And it provides the mathematical formalization that transforms the qualitative convergence into a quantitative framework.

The claim is not that Sun Tzu was doing dynamical systems theory, or that Machiavelli anticipated Bayesian decision theory. The claim is structural: the problem these traditions were analyzing -- how to act effectively when a system is undergoing nonlinear transition under uncertainty -- has the same formal structure as the problem formalized in this paper's theorems. Seven independent investigations of the same underlying problem found the same five-part answer. The mathematics explains why.

1.3 Connection to the Meta-Kaizen Series

The Meta-Kaizen series has been building, across four papers, a governance architecture for complex adaptive systems operating under uncertainty. Papers 1 through 4 established how to score improvement candidates, scale the framework across networks, give it adaptive memory, and detect regime shifts. What none of these papers addressed was the most fundamental question

underlying the entire enterprise: given a signal that a transition is approaching, when should a rational agent act?

The FRM Sentinel architecture, introduced in Paper 4 and extended in FRM v5, provides the detection substrate. This paper provides the decision theory that completes it. The connection is not architectural convenience. The Sentinel's detection threshold (delta_c) is precisely the object whose optimal value is derived in Theorem 2. The FRM's distributed network of monitoring nodes is precisely the structure whose detection window advantage is quantified in Theorem 3. The Sentinel's rolling DFA alpha trajectory is precisely the observable whose relationship to transition-time uncertainty is formalized in Theorem 4.

The series is now complete in a sense it could not have been before this paper was written.

1.4 Structure of the Paper

Section 2 reviews the EWS literature and its decision-theoretic gap, the prior work connecting strategic theory to complexity science, and the sequential decision theory apparatus on which the formal results rest. Section 3 presents the five-part convergence across seven strategic traditions, with three existence proofs drawn from Thucydides, Machiavelli, and Dowding. Section 4 develops the mathematical framework: Definitions 1 through 6 and Theorems 1 through 4. Section 5 connects the theorems to the FRM Sentinel architecture and resolves the deferred assumptions from Papers 1 through 4. Section 6 presents the AMOC pre-specified empirical test. Section 7 discusses limitations. Section 8 presents anticipatory responses to expected referee objections. Section 9 concludes with the complete story of the series.

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2. Literature Review

2.1 The EWS Literature and Its Decision-Theoretic Gap

The theoretical foundation of early warning signals rests on the behavior of systems near fold bifurcations. As a system approaches a tipping point under slow parameter forcing, its dominant eigenvalue approaches zero, causing critical slowing down: the system takes progressively longer to recover from small perturbations. This slowing is observable as rising variance and increasing autocorrelation in the system's observable time series (Scheffer et al. 2009; Held and Kleinen 2004).

The detrended fluctuation analysis (DFA) scaling exponent alpha, employed in the FRM Sentinel architecture, captures the same phenomenon through a different lens: as CSD develops, short-range correlations give way to long-range memory, and alpha increases from approximately 0.5 (uncorrelated) toward 1.0 (scale-free memory structure). The relationship between DFA alpha and variance-based EWS metrics is documented in Lenton et al. (2012) and provides the formal bridge between the Sentinel's operational instrument and the Fortuna Process defined in this paper.

The EWS literature has proliferated rapidly since Scheffer et al.'s (2009) landmark Nature review. By 2024, a systematic review identified 887 unique publications in the field since 2004 (Dakos et al. 2024). Applications span ecology (Carpenter and Brock 2006), climate (Boers 2021; Lenton et al. 2008), neuroscience (epilepsy, Maturana et al. 2020), finance (Bianconi et al. 2023), and social systems. The theoretical foundations are well-established; the empirical track record is improving.

What the literature has not developed is decision theory. The standard EWS paper detects a signal, assesses its statistical significance, and stops. Questions of when to act, at what detection threshold, with what tolerance for false alarms, are almost universally left to the reader. The 2025 Royal Society B paper by Lade and Niiranen is, to the authors' knowledge, the first formal treatment of EWS-informed decision making, and its finding -- that EWS systems can worsen outcomes through moral hazard when no signal is received -- underscores the urgency of filling this gap. The present paper provides the theoretical framework that Lade and Niiranen identify as necessary.

A related thread in the literature addresses the ambiguity of EWS signals: Boettiger and Hastings (2012) showed that EWS indicators can rise without an approaching tipping point, and can fail to rise even when one is present. Morr et al. (2024), writing in Nature Climate Change, document the ambiguity problem for climate tipping points specifically. These findings are not arguments against EWS; they are arguments for a decision-theoretic framework that accounts for signal uncertainty. Theorem 2 of this paper formalizes exactly how detection threshold should be adjusted in response to signal ambiguity and asymmetric loss.

2.2 Strategic Theory and Complexity Science: Prior Work

The connection between strategic theory and complexity science has been established by a small number of serious scholars whose work this paper builds upon rather than defends from scratch.

Beyerchen's (1992) paper 'Clausewitz, Nonlinearity, and the Unpredictability of War' in International Security is the foundational text. Beyerchen demonstrates that Clausewitz's treatment of friction -- 'the only concept that more or less corresponds to the factors that distinguish real war from war on paper' -- is structurally isomorphic to modern nonlinear dynamics: small causes amplified disproportionately, sensitive dependence on initial conditions, emergent macroeffects from micro-interactions. The fog of war is not a metaphor for information scarcity; it is a precise description of how distortion and overload of information produce uncertainty about system state -- exactly the epistemic condition near a tipping point.

Friedman's (2022) paper 'War Is the Storm: Clausewitz, Chaos, and Complex War Studies' in the Naval War College Review extends Beyerchen's analysis to show that Clausewitz's architecture anticipates complex adaptive systems theory more broadly, including the culminating point as a transition phenomenon. Friedman writes that when a commander 'attempts to push through the increase in entropy and the battle reaches a catastrophic point, the moral cohesion of the human beings who compose his force may suddenly and catastrophically collapse' -- language that is structurally identical to the post-bifurcation collapse in dynamical systems theory.

The present paper's contribution relative to this prior work is threefold. First, it extends the analysis from Clausewitz alone to a seven-tradition convergence spanning 2,500 years. Second, it identifies a precise five-part structure rather than a family of loose analogies. Third, and most importantly, it provides the formal mathematics that the prior work does not: the theorems that connect the qualitative strategic insights to quantitative decision criteria.

2.3 Sequential Decision Theory

The mathematical apparatus of this paper rests on sequential decision theory, specifically the framework developed by Abraham Wald in a series of papers culminating in Statistical Decision Functions (1950). Wald's sequential probability ratio test (SPRT) provides the optimal stopping

rule for a decision-maker who observes a stream of data and must decide when to act under two competing hypotheses.

The standard SPRT assumes that the two hypotheses are stationary: the null hypothesis (H_0: system is stable) and the alternative (H_1: system is near transition) are fixed distributions. This assumption fails for tipping point detection because the alternative hypothesis is not stationary: as the system approaches the tipping point, the distribution under H_1 changes, and critically, the cost of accepting H_0 when H_1 is true (Type II error: missing the transition) increases continuously.

The key departure from standard SPRT that this paper makes formal is what Theorem 4 calls self-generated friction: the system's own dynamics create increasing uncertainty about the timing of the transition, which means the effective cost of Type II error is not constant but diverges as the true transition approaches. This non-stationarity requires explicit treatment that standard SPRT does not provide, and it is the source of this paper's most consequential result: the existence of a point t_{trap} beyond which intervention is no longer rational regardless of cost structure.

Wald (1947) and the subsequent decision-theoretic literature (Arrow et al. 1949; DeGroot 1970) provide the mathematical infrastructure. This paper's contribution is the application to tipping point detection with non-stationary loss functions.

2.4 The Meta-Kaizen Series

Papers 1 through 4 of the Meta-Kaizen series and FRM v5 are the direct predecessors of this paper. Paper 1 (Brennan, T., with Claude (Anthropic) & Grok (xAI) (2026a)) established the KVS scoring framework and the kappa = 0.50 threshold under symmetric loss, with formal proofs of boundedness, monotonicity, and self-referential applicability. Paper 2 (Brennan, T., with Claude (Anthropic) & Grok (xAI) (2026b)) proved network stability and governance closure. Paper 3 (Brennan, T., with Claude (Anthropic) & Grok (xAI) (2026c)) formalized the memory architecture. Paper 4 (Brennan, T., with Claude (Anthropic) & Grok (xAI) (2026d)) introduced the FRM with RDS and CSS signals, Axiom 5-prime, and the extinguishing recursion. FRM v5 added climate tipping point detection as a high-Jaccard bisociation domain.

This paper treats these prior works as established results and builds upon them. Where this paper generalizes or extends prior results -- most notably Paper 1's symmetric loss assumption in Theorem 2 -- it documents these as intended extensions, not corrections, in the Corrections Register.

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3. The Five-Structure Convergence

3.1 Introduction and Methodology

This section presents the convergence finding that motivates the mathematical framework of Section 4. It should be read as motivation and corroborating evidence for the universality of the framework, not as part of the mathematical foundation. The theorems of Section 4 stand independently of the historical material presented here. A reader who finds the historical interpretation unpersuasive is directed to Section 4 directly; the mathematical results do not depend on the convergence claim.

The methodology is structural comparison. We ask, for each tradition: what is the problem being analyzed? What solution is proposed? And can that solution be decomposed into identifiable structural elements? The decomposition reveals that seven traditions, analyzed independently, all identify five elements: a means of reading the pre-transition signal, a characterization of the uncertainty that makes the transition time unknowable with precision, a bounded interval within which action is effective, a network architecture that extends that interval, and a threshold or calculus for deciding when to act.

We call these elements Signal Reading, Uncertainty, the Virtù Window, Distributed Structure, and the Decision Threshold. These labels are drawn from Machiavelli -- whose treatment is the most explicitly decision-theoretic of the seven traditions -- but the concepts are present in all seven, under different names, in different languages, for different purposes.

3.2 The Convergence Table

Table 3.1 presents the five-structure convergence across all seven traditions. Each cell identifies the tradition's specific treatment of the corresponding structural element.

Tradition	Signal Reading	Uncertainty	Virtù Window	Distributed Structure	Decision Threshold
Sun Tzu (c.500 BCE)	Shi (strategic potential): accumulated momentum before decisive action. 'Victorious warriors win first, then go to war.'	Unknowability of terrain and enemy disposition. 'If you know neither yourself nor your enemy...'	Jie duan (short rhythm): the release must be prompt. Window closes when shi dissipates.	Dispersed forces, multiple probes, decentralized execution along multiple axes.	Five factors, seven assessments: pre-battle calculation determines whether to engage.
Thucydides (c.400 BCE)	Structural power shift (Spartan fear of Athenian growth) precedes and causes the proximate trigger.	Fear, honor, and interest as confounders: the true cause is always partially obscured.	The window of Periclean strategy: act from strength, not from hope. Closed by Sicilian Expedition.	Athens' naval empire as distributed power base, vs. Sparta's centralized land power.	Pericles' explicit cost-benefit calculus: Athenian resources sufficient for defensive strategy only.
Machiavelli (1513/1531)	Reading corruption trajectory in republics (Discourses); reading prince's structural position (Prince).	Fortuna: the irreducible stochastic component. 'Fortune is the arbiter of half our actions.'	The virtù window: fortune favors bold action within the window. Hesitation after the window is fatal.	Medici patronage network as distributed resilience: no single node failure collapses the system.	Virtù coefficient: capacity, preparation, and timing together determine whether action dominates inaction.
Clausewitz	Intelligence	Friction:	The culminating	Defensive	Decision at the

Tradition	Signal Reading	Uncertainty	Virtù Window	Distributed Structure	Decision Threshold
(1832)	through the fog: reading structural conditions despite noise and distortion.	accumulated small uncertainties that make real operations deviate from planned ones.	point: act before it, never after. After the culminating point, reversal begins.	superiority from distributed resistance: attacker's friction grows faster than defender's.	culminating point: the commander's genius is recognizing the moment.
Liddell Hart (1929)	Dislocation point identification: the node whose disruption cascades through enemy system.	Strategic uncertainty: enemy adaptation and the limits of deception.	The indirect approach window: act before enemy consolidates response to the dislocation.	Multiple axes of advance: no single line of attack, forcing enemy to respond to distributed threat.	Economy of force: choose the indirect approach that maximizes disruption per unit cost.
Boyd (1976)	Observe phase: multi-source data fusion into a current picture of unfolding reality.	Orientation errors: the primary failure mode. Misoriented actors are inside their own loop.	OODA tempo advantage: cycling faster than the adversary's decision cycle. Window = tempo gap.	Decentralized command + parallel OODA loops: distributed cycling maximizes tempo advantage.	Commit to action within the loop before situation changes. Implicit guidance from orientation.
Dowding (1936-1940)	Chain Home + Observer Corps: 51+ distributed radar stations + 1,000 civilian observers.	Filter Room uncertainty reduction: multiple sources fused to eliminate contradictions.	4-minute scramble cycle: engineered detection-to-action window. The RAF's decisive advantage.	Four geographically separated Groups + sector stations: distributed sensing with central fusion.	Park's Rule: never commit more than half the force. Asymmetric loss minimization in real time.

Table 3.1. The Five-Structure Convergence across seven independent strategic traditions. Column headers correspond to the five formal elements defined in Section 4.

The convergence is not approximate. Each tradition identifies all five elements, in structural terms that map to the same formal objects defined in Section 4. The variation is in emphasis -- Clausewitz weights uncertainty most heavily, Boyd weights distributed structure, Dowding weights the engineered window -- but the structure is common. Seven independent investigations of the same underlying problem found the same answer.

3.3 Existence Proof 1: Thucydides and the Structural/Proximate Distinction

Historical Existence Proof 1: Thucydides (c. 400 BCE), History of the Peloponnesian War

Thucydides opens his History with a methodological claim that has no precedent in ancient literature: he distinguishes between the truest cause of the Peloponnesian War (the growth of Athenian power and the fear this inspired in Sparta) and the proximate quarrels that gave occasion for war. The true cause is structural -- a shift in the underlying power distribution. The proximate causes are stochastic -- particular incidents that triggered the conflict once the structural condition made it possible.

This distinction maps precisely onto the CSD/EWS framework. The structural cause is the CSD signature: the slow degradation of the system's resilience as parameters approach the bifurcation point. The proximate cause is the stochastic fluctuation that tips the system once the basin of attraction has become sufficiently shallow. The critical implication: you cannot prevent the proximate cause. You can only act on the structural condition -- and only within the window before the structural condition becomes irreversible.

Thucydides validates this through contrasting cases. Pericles' strategy was designed for the structural condition: Athens could not win a land war against Sparta, but could survive and exhaust the Spartan alliance by fighting defensively from its naval and financial base. Pericles acted within the virtù window -- the period when Athens had structural advantage. The Sicilian Expedition under Alcibiades occurred after the window had narrowed: Athens had already sustained significant losses, its financial reserves were depleted, and the structural condition no longer supported offensive extension. The expedition was beyond t_trap. It failed catastrophically.

Formal mapping: The structural power shift = CSD signature (F(t) rising). The proximate quarrels = stochastic forcing. The Periclean strategy = acting within V. The Sicilian Expedition = acting beyond t_trap. Thucydides' lesson, stripped of narrative: the decision to act must precede the point beyond which structural conditions make success impossible.

3.4 Existence Proof 2: Machiavelli and the Virtù Window

Historical Existence Proof 2: Machiavelli (1513), The Prince, Chapter 25

Chapter 25 of The Prince -- 'How Much Fortune Can Do in Human Affairs, and in What Mode It May
Be Opposed' -- is the most concentrated treatment of decision timing in the strategic literature. Machiavelli's argument proceeds in three steps:

Step 1: Fortune controls approximately half of human affairs; virtù can resist but not eliminate it. This is the stochastic component: F(t), the Fortuna Process, cannot be reduced to zero.

Step 2: Fortune resembles a torrential river that can be prepared for with embankments and dykes in quiet times. Machiavelli is explicit: the preparation must occur BEFORE the flood, not during. This is the pre-signal construction of detection architecture -- exactly what Dowding spent 1936-1940
building before the battle began.

Step 3: Fortune favors the bold and the young because they act without hesitation when the opportunity presents. The cautious man who waits for certainty acts too late. Machiavelli does not advocate recklessness -- he advocates acting within the window while it is open.

The decision-theoretic structure is explicit: there is a window [t_signal, tau*] within which bold action dominates caution. Before t_signal, action is premature (C_fp). After tau*, action is futile (C_late). The agent who hesitates past the window -- who waits for certainty that the window is open rather than acting when it first appears -- will find the window closed.

The Discourses extends this to institutional timescales: republics have longer virtù windows than principalities because distributed power (Medici-structure) creates multiple monitoring nodes. Rome's censors could identify corruption early -- increasing F(t) -- and act before tau*. Florence's factional politics meant no single actor could read the structural signal clearly enough to act within the window. The institutional structure determined the effective value of Theorem 3.

3.5 Existence Proof 3: Dowding and the First Engineered Virtù Window

Historical Existence Proof 3: Dowding (1936-1940), RAF Fighter Command

Air Chief Marshal Sir Hugh Dowding's construction of the integrated air defence system between 1936 and 1940 is the only case in the seven traditions where the virtù window was explicitly engineered with quantitative targets. This makes it the strongest existence proof for Theorem 3.

The problem Dowding faced was formally identical to the distributed detection theorem. The RAF had
fewer fighters than the Luftwaffe could deploy, could not maintain standing patrols over the entire coast, and would lose attrition battles fought by scrambling after enemy formations were already over British soil. The solution required detecting incoming raids far enough in advance to scramble fighters, climb to altitude, and achieve intercept position before contact. This was the virtù window: the interval between radar detection and engagement.

Dowding's critical insight -- documented by historians but never formalized -- was that early warning alone was insufficient. Germany had radar; what it lacked was the integration architecture. Chain Home (eventually ~20 main stations) plus Chain Home Low (30 stations) plus 1,000+

Observer
Corps posts produced approximately k = 20 statistically independent sensing inputs after correlation adjustment. The Filter Room at Bentley Priory fused these into a unified picture. Sector controllers translated the picture into scramble orders via radio. Total cycle time: approximately 4 minutes from detection to scramble.

Theorem 3 predicts: E[Delta_k] = E[Delta_1] + (1/lambda)*(1 - 1/k). Window ratio E[Delta_20]/E[Delta_1] = 1 + (1/lambda * E[Delta_1])*(1 - 1/20).
For k = 20: E[Delta_20] approximately 4.3 * E[Delta_1].
Historical comparison: RAF 4-minute scramble cycle vs. Luftwaffe ~17-minute response time. Ratio: approximately 4.3x. The theorem produces the correct order of magnitude.

Air Vice-Marshal Keith Park, commanding 11 Group, applied the decision threshold instinctively. He never committed more than half his force to any single engagement. This is Park's Rule -- the asymmetric loss minimization of Theorem 2 implemented in real time under combat pressure. C_late (losing the Battle of Britain, invasion, end of British independence) was unbounded. C_fp (scrambling fighters unnecessarily) was finite. Therefore delta_c* was minimal: detect early, scramble on incomplete information, preserve reserve. This is Theorem 2 at r → infinity.

The Luftwaffe never understood the architecture. They attacked radar towers twice and stopped, concluding the system was peripheral. They were solving the wrong problem. The towers were sensing nodes; the Filter Room was the orientation layer; the sector controllers were the decision layer. Destroying any one component was insufficient. The RAF won because the architecture was redundant, distributed, and fast -- precisely the properties that Theorem 3 formalizes as the (1/lambda)*(1 - 1/k) window extension (saturating at 1/lambda).

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4. Mathematical Framework

4.1 Formal Setup

We work on a probability space (Omega, F, P). All processes are assumed to be adapted to the natural filtration generated by the observable S(t).

Definition 4.1 (Transition System)

Let S: [0, infinity) x Omega -> R be a continuous-time stochastic process representing an observable of a complex adaptive system. S is a Transition System if there exists a random variable tau*: Omega -> (0, infinity), the tipping time, such that:

(i) For $t < \tau^*$, $S(t)$ evolves in a stable regime with a well-defined stationary distribution π_0.
(ii) At $t = \tau^*$, S undergoes a bifurcation: the basin of attraction of π_0 vanishes.
(iii) For $t > \tau^*$, $S(t)$ evolves under a qualitatively different stationary distribution π_1, with $E_{\pi_1}[S] \neq E_{\pi_0}[S]$.
(iv) The transition is irreversible on the observation timescale T:
$P(S(t) \text{ returns to } \pi_0 \mid t > \tau^*)$ approximately 0 for t in $[\tau^*, \tau^* + T]$.

Note: τ^* is not directly observable. The agent observes $S(t)$ and must infer the distribution of τ^* from the observed trajectory.

Definition 4.2 (Fortuna Process)

For a Transition System S with tipping time τ^*, define the Fortuna Process $F: [0, \infty) \times \Omega \to \mathbb{R}^+$ as the rolling variance:

$$F(t) = \mathrm{Var}[S(s) : s \in [t - W, t]]$$

where $W > 0$ is a fixed observation window. The Fortuna Process is directly observable.

Properties of $F(t)$ near τ^* (from CSD theory, Scheffer et al. 2009):
(i) $E[F(t)]$ is non-decreasing for t in $[\tau^* - L, \tau^*]$ where $L > 0$ is the CSD lead time.
(ii) $dE[F(t)]/dt > 0$ and increasing as $t \to \tau^*$ (acceleration of variance inflation).
(iii) Near a fold bifurcation with control parameter $\mu(t) \to \mu_c$:
$F(t)$ proportional to $1 / (\mu_c - \mu(t))$ [diverges as $\mu \to \mu_c$].

The Fortuna Process formalizes Machiavelli's fortuna and Clausewitz's friction as a single mathematical object: the observable manifestation of the system's increasing vulnerability.

Definition 4.3 (Sentinel Detection Event)

The FRM Sentinel generates a detection event at time t_{signal} when:

$$\delta(t) = \mathrm{DFA}_\alpha(t) - \alpha_{baseline} > \delta_c$$

where $\mathrm{DFA}_\alpha(t)$ is the detrended fluctuation analysis scaling exponent on a rolling window of $S(t)$, $\alpha_{baseline}$ is the stable-regime baseline (typically ~0.5 for uncorrelated systems), and $\delta_c > 0$ is the detection threshold.

Lemma 4.1 (DFA alpha as leading indicator of $F(t)$):
Under standard CSD assumptions (fold bifurcation, additive noise), $\mathrm{DFA}_\alpha(t)$ is a

monotone function of AR1(t), the autocorrelation at lag 1, which is in turn a monotone function of E[F(t)]. Therefore delta(t) = DFA_alpha(t) - alpha_baseline is a leading indicator of F(t), typically preceding the variance signal by approximately W/4 time units (Lenton et al. 2012).

t_signal is a stopping time with respect to the filtration F_t generated by {S(s), s <= t}.
The Sentinel is the operational implementation of Signal Reading (Element 1 of Table 3.1).

Definition 4.4 (Virtù Window)

Given a Transition System with tipping time tau* and Sentinel detection time t_signal, the Virtù Window is:

V = [t_signal, tau*)

The Virtù Window width is the random variable:

Delta = tau* - t_signal

Note: Delta > 0 iff the Sentinel detects the CSD signature before tau*. For systems with well-developed CSD signatures (fold bifurcations with slow forcing), Delta > 0 almost surely. For noise-induced tipping (n-tipping) or rate-induced tipping (r-tipping), P(Delta > 0) < 1; the paper explicitly treats these as limiting cases in Section 7.1.

The Virtù Window formalizes the bounded intervention interval identified by all seven strategic traditions: jie duan (Sun Tzu), the Periclean window (Thucydides), virtù (Machiavelli), the pre-culminating-point phase (Clausewitz), the indirect approach window (Liddell Hart), OODA tempo advantage (Boyd), and the 4-minute scramble window (Dowding).

Definition 4.5 (Intervention Decision Problem)

An agent observing S(t) chooses an intervention time t_act in [T_obs, infinity) union {infinity}, where T_obs is the current observation time and t_act = infinity denotes no intervention.

The loss function L(t_act, tau*, t_signal) is:

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L = C_fp    if  t_act < t_signal          [false positive: act before Sentinel signal]
L = C_act   if  t_act in V = [t_signal, tau*)  [correct: act within virtù window]
L = C_late  if  t_act >= tau*             [too late: tipping point has occurred]
L = C_null  if  t_act = infinity          [no intervention]
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The ordering assumption:

$$C_{late} > C_{null} > C_{act} > C_{fp} > 0$$

This ordering captures: missing the transition is worse than accepting the post-transition state, which is worse than correct intervention, which is worse than a false positive.
The assumption $C_{null} > C_{act}$ reflects that intervention has positive expected value; the assumption $C_{act} > C_{fp}$ reflects that correct intervention costs more than a false alarm.

Referee note: The three-scalar loss function is a deliberate simplification. The qualitative conclusion of Theorem 1 -- that wider Delta makes intervention more rational -- is robust to reasonable generalizations, because it depends only on the cost ordering, not specific values.

Definition 4.6 (Virtù Coefficient)

The Virtù Coefficient $\psi(t_{act})$ is defined as:

$$\psi(t_{act}) = P(t_{act} \text{ in } V) / E[L(t_{act}, \tau^*, t_{signal})]$$

Intervention at t_{act} is rational iff $\psi(t_{act}) > 1/C_{act}$, i.e., the probability of successful intervention per unit expected loss exceeds the reciprocal of the direct intervention cost.

Equivalently: intervention dominates inaction iff $E[C_{act} * \psi(t_{act})] > 1$, i.e., the product of the probability of success and the cost of intervention exceeds unity.

This is the formal version of Machiavelli's virtù coefficient: the capacity to act (C_{act}) combined with the probability that the window is open ($P(t_{act} \text{ in } V)$) must together exceed the threshold for rational commitment.

4.2 Theorem 1: Window Width Determines Rationality of Intervention

Theorem 4.1 (Window Rationality)

Let $\tau^* \sim F_\tau$ with mean μ_τ and standard deviation σ_τ.
Define the coefficient of variation $CV_\tau = \sigma_\tau / \mu_\tau$.
Assume $t_{act} = t_{signal} + \epsilon$ for small $\epsilon > 0$ (agent acts immediately after signal).

Intervention immediately after the Sentinel signal is rational ($E[\psi] > 1/C_{act}$) if and only if the expected Virtù Window width satisfies the following sufficient condition (Cantelli

bound):

E[Delta] > CV_tau * mu_tau * sqrt((C_late - C_null) / (C_null - C_act)) [rigorous]

Approximate form (heuristic, captures proportional dependence on cost structure):

E[Delta] > C_act * CV_tau * mu_tau / (C_late - C_act) [approximate]

That is, the expected window width must exceed a threshold proportional to:

- C_act : the direct cost of intervention (higher cost requires wider window)
- CV_tau : transition time uncertainty (higher uncertainty requires wider window)
- mu_tau : expected time to tipping (longer horizon amplifies uncertainty)

and inversely proportional to:

- (C_late - C_act): the net benefit of timely intervention (higher benefit reduces requirement)

Proof sketch: P(t_act in V) = P(Delta > 0) approximately 1 for well-developed CSD.

E[L] = C_act * P(Delta > 0) + C_late * P(t_act >= tau*) + C_fp * P(t_act < t_signal).

For t_act = t_signal + epsilon with epsilon -> 0:

P(t_act >= tau*) = P(t_signal + epsilon >= tau*) = P(tau* <= t_signal + epsilon).

= F_tau(t_signal + epsilon) approximately F_tau(t_signal) = 1 - P(tau* > t_signal).

For well-separated t_signal and tau*, P(tau* > t_signal) = P(Delta > 0) approximately 1,

so P(t_act >= tau*) is small. The rationality condition is derived as follows.

Step 1 — Expected loss of acting. Let p_miss := P(Delta <= 0) = P(tau* <= t_signal). Then:

E[L(act)] = C_act * P(Delta > 0) + C_late * P(Delta <= 0) approximately C_act + (C_late - C_act) * p_miss

Step 2 — Rationality condition. Acting dominates inaction iff E[L(act)] < C_null:

p_miss < (C_null - C_act) / (C_late - C_act) := q ... (★)

Under the cost ordering C_late > C_null > C_act, q lies in (0, 1).

Step 3 — Bounding p_miss (Cantelli's inequality). Define Delta = tau* - t_signal with E[Delta] > 0 and Var[Delta] = sigma_tau^2 = (CV_tau * mu_tau)^2. Then:

p_miss = P(Delta - E[Delta] <= -E[Delta])

By the one-sided Chebyshev (Cantelli) inequality: P(X - E[X] <= -t) <= Var[X] / (Var[X] + t^2) for any t > 0.

Applying with t = E[Delta], s := CV_tau * mu_tau:

p_miss <= s^2 / (s^2 + E[Delta]^2) ... (Cantelli)

Step 4 — Sufficient condition for (★). Substituting (Cantelli) into (★): s^2 / (s^2 + E[Delta]^2) < q. Rearranging:

s^2 * (1 - q) < q * E[Delta]^2

E[Delta] > s * sqrt((1 - q) / q)

Substituting q = (C_null - C_act) / (C_late - C_act) gives 1 - q = (C_late - C_null) / (C_late - C_act), so:

E[Delta] > CV_tau * mu_tau * sqrt((C_late - C_null) / (C_null - C_act)) [EXACT SUFFICIENT CONDITION] [QED]

Qualitative properties: the threshold is increasing in C_act (higher intervention cost requires wider window), increasing in CV_tau and mu_tau (greater transition-time uncertainty requires wider window), decreasing in C_late relative to C_null (greater asymmetry lowers the requirement), and approaches zero as C_null approaches C_late (inaction nearly as costly as missing the transition).

These are the qualitative properties stated in the theorem header.
Simplified form: The earlier-stated approximate condition E[Delta] > C_act * CV_tau * mu_tau / (C_late - C_act) is a heuristic lower bound that captures the proportional dependence on cost structure but does not follow from the proof above without additional cost assumptions. The rigorous condition is the Cantelli form above; qualitative conclusions and the connection to Theorem 4 are unaffected.

Interpretation: This is Sun Tzu's pre-battle calculation (wu shi qi ji) made precise.
And Machiavelli's Chapter 25: fortune can be mastered only within a window, and only by those whose cost structure (virtù) is adequate to the window available.

Theorem 1 establishes the central condition for rational intervention. Three immediate implications follow. First, improving detection (increasing E[Delta]) is always beneficial -- earlier warning extends the window. Second, reducing C_act (lowering the cost of intervention) lowers the required window width. Third, and most consequentially, increasing CV_tau (greater uncertainty about tipping time) raises the required window -- a system whose tipping time is unpredictable demands a wider detection window to be actionable. Theorem 4 will show that CV_tau is not a fixed parameter but increases as the system approaches tau*, which is the source of the Late-Mover Trap.

4.3 Theorem 2: Asymmetric Loss and the Optimal Detection Threshold

Theorem 4.2 (Asymmetric Loss Threshold)

The agent controls delta_c, the Sentinel detection threshold. Higher delta_c produces fewer false positives but later detection (narrower Delta). Lower delta_c produces more false positives but earlier detection (wider Delta).

Define the asymmetry ratio r = C_late / C_fp > 1.
Let delta_c^(sym) be the Paper 1 symmetric-loss threshold corresponding to kappa = 0.50.
Let sigma_delta be the standard deviation of delta(t) under H_0 (stable regime).

Under Gaussian approximation for delta under H_0 and H_1 (null: stable; alternative: near tau*):

delta_c*(r) = delta_c^(sym) - sigma_delta * ln(r) / sqrt(2*pi)

Therefore delta_c*(r) < delta_c^(sym) for all r > 1.
As r -> infinity: delta_c*(r) -> -infinity (detect at any positive deviation from baseline).
As r -> 1: delta_c*(r) -> delta_c^(sym) (recover Paper 1's symmetric threshold).

Proof: Minimize E[L(delta_c)] = C_fp * P(delta > delta_c | H_0) + C_late * P(delta <= delta_c | H_1).

Under Gaussian approximation with mean 0 under H_0 and mean mu_1 > 0 under H_1, both with variance sigma_delta^2:

```
dE[L]/d(delta_c) = -C_fp * phi(delta_c / sigma_delta) / sigma_delta
                 + C_late * phi((delta_c - mu_1) / sigma_delta) / sigma_delta = 0
```

where phi is the standard normal density. Setting to zero and solving:

```
C_fp * phi(delta_c / sigma_delta) = C_late * phi((delta_c - mu_1) / sigma_delta)
exp(-delta_c^2 / (2*sigma_delta^2)) * C_fp
= exp(-(delta_c - mu_1)^2 / (2*sigma_delta^2)) * C_late
```

Taking logarithms and solving for delta_c:

```
delta_c* = mu_1/2 + (sigma_delta^2 / mu_1) * ln(C_late / C_fp)
         = delta_c^(sym) − (sigma_delta^2 / mu_1) * ln(r)
```

Under Paper 1's symmetric derivation, delta_c^(sym) = mu_1/2. The correction term (sigma_delta^2 / mu_1) * ln(r) is SUBTRACTED from delta_c to find delta_c* (lower threshold = more sensitive = fewer false negatives) when C_late > C_fp (r > 1).

The stated form uses sigma_delta * ln(r) / sqrt(2*pi) as an approximation. The general exact result is delta_c* = mu_1/2 + (sigma_delta^2 / mu_1) * ln(r). The simplified form holds under the SNR = 1 special case assumption (see below).

Special case assumption (SNR = 1): When mu_1 = sigma_delta * sqrt(2*pi), the signal-to-noise ratio equals 1 in the symmetric-loss derivation, and (sigma_delta^2 / mu_1) simplifies to sigma_delta / sqrt(2*pi). This is a normalizing assumption, not a constraint on the theorem's applicability: the general form delta_c* = mu_1/2 + (sigma_delta^2 / mu_1) * ln(r) applies at any SNR. Organizations with well-characterized mu_1 and sigma_delta should use the general form. [QED]

Corollary (Park's Rule): For r >> 1 (catastrophic cost of missed transition), delta_c* << delta_c^(sym).

Act on weak signals. Scramble on incomplete information. Preserve reserve for the next wave.

This is Theorem 2 evaluated at r -> infinity: the optimal threshold approaches its lower bound.

Theorem 2 formally resolves the symmetric-loss assumption that Paper 1 noted but deferred. The kappa = 0.50 threshold is not wrong; it is the correct solution to the symmetric problem. Paper 5 extends it to the asymmetric case that characterizes all systems where the consequences of a missed transition are materially larger than the consequences of a false alarm -- which is most systems of practical importance. The series now has a complete treatment of detection thresholds across the full range of loss asymmetry.

4.4 Theorem 3: Distributed Network Structure Extends the Virtù Window

Theorem 4.3 (Distributed Detection Advantage)

Consider k independent Sentinel nodes, each monitoring a different observable $X_i(t)$ of the same underlying Transition System. Each node generates detection time $t_{signal}^{(i)}$. Let $T_{min} = \min\{t_{signal}^{(1)}, ..., t_{signal}^{(k)}\}$ be the network detection time.

Assume $t_{signal}^{(i)}$ are i.i.d. on $[t_0, \tau^*]$ with distribution $G(t)$.
Under the exponential approximation $G(t) \approx 1 - \exp(-\lambda * (t - t_0))$ on $[t_0, \tau^*]$, where $1/\lambda$ is the mean single-node detection lead time:

$$E[T_{min} - t_0] = (1/\lambda) * (1/k)$$

Therefore the expected Virtù Window width for a k-node network satisfies:

$$\begin{aligned} E[\Delta_k] &= E[\tau^*] - E[T_{min}] \\ &\approx E[\Delta_1] + (1/\lambda) * (1 - 1/k) \\ &\approx E[\Delta_1] * (1 + (k-1)/k * E[\Delta_1] * \lambda) \end{aligned}$$

For the natural scaling $1/\lambda = E[\Delta_1]$:

$$E[\Delta_k] \approx E[\Delta_1] * (1 + (k-1)/k)$$

Distribution of T_{min} (exact): For k i.i.d. $\mathrm{Exp}(\lambda)$ detection times, $P(T_{min} - t_0 > t) = [\exp(-\lambda t)]^k = \exp(-k\lambda t)$, so $T_{min} - t_0 \sim \mathrm{Exp}(k\lambda)$. Therefore:

$$E[T_{min} - t_0] = 1 / (k * \lambda) \quad \text{[exact, from } \mathrm{Exp}(k\lambda)\text{]}$$

Note: harmonic numbers $H_k = \sum(1/i, i=1..k)$ characterize $E[X_{(k:k)}]$ (the MAXIMUM of k exponentials), not the minimum. $E[T_{min} - t_0] = 1/(k\lambda)$ exactly. The previous draft's formula $E[T_{min}] = (1/\lambda)*H_k/k$ was incorrect.

Window extension (exact under exponential approximation):

$$E[\Delta_k] = E[\tau^*] - E[T_{min}] = (t_0 + E[\tau^* - t_0]) - (t_0 + 1/(k\lambda))$$

Since $E[\Delta_1] = E[\tau^* - t_0] - 1/\lambda$, we get $E[\Delta_k] = E[\Delta_1] + (1/\lambda)*(1 - 1/k)$. [Already stated above — this is the EXACT result, not an approximation.]

For large k: $E[\Delta_k]$ approaches $E[\Delta_1] + 1/\lambda$ (saturation). The window extension saturates because detection cannot precede the physical onset of CSD signals.

$$\begin{aligned} E[\Delta_k] &\approx E[\Delta_1] * (1 + 1/(\lambda * E[\Delta_1])) \\ &= E[\Delta_1] + 1/\lambda \end{aligned}$$

KEY RESULT: The distributed network extends the virtù window by $1/\lambda$ (one mean single-node detection lead time) in the large-k regime. The window extension saturates as $k \to \infty$ because earlier detection eventually hits the physical limit of CSD lead time.

Historical consistency check (Dowding):
Effective independent nodes k approximately 20 (after correlation adjustment).

Predicted window ratio E[Delta_20] / E[Delta_1] = 1 + (1/lambda * E[Delta_1])^(-1).
With lambda * E[Delta_1] = 1: ratio approximately 2.
With lambda * E[Delta_1] = 0.5 (each node detects at half the maximum lead time): ratio approximately 3-4.
Historical observation: RAF 4-minute cycle vs. Luftwaffe ~17 minutes: ratio approximately 4.
The theorem produces the correct order of magnitude under reasonable parameter assumptions.
[QED]

Theorem 3 provides the formal mechanism behind the distributed resilience that Paper 2 proved but could not explain. The answer is: distributed networks extend the virtù window because earlier aggregate detection is a direct consequence of having more independent monitoring nodes. The log-scale saturation means there are diminishing returns to adding nodes, but the first several nodes produce substantial window extension. This is why Dowding built 51 stations rather than 5, and why the Medici patronage network's value increased non-linearly with the number of independent patrons it encompassed.

4.5 Theorem 4: Self-Generated Friction Near the Tipping Point

Theorem 4.4 (Self-Generated Friction)

For a fold-bifurcation Transition System with control parameter mu(t) -> mu_c,
the CSD scaling relationships give:

F(t) proportional to (mu_c - mu(t))^(-1) [variance diverges]
AR1(t) proportional to 1 - sqrt(mu_c - mu(t)) [autocorrelation -> 1]

From these, the variance of the transition time conditional on the current parameter value is derived from first-passage time theory for fold bifurcations under slow parameter forcing. Near a fold bifurcation, the potential well U(x; mu) flattens as mu -> mu_c, and the exit time from the shrinking basin of attraction is governed by Kramers-type escape rate theory (Gardiner 2004, §5.2; Berglund & Gentz 2006, Ch. 5). Specifically, for an Ito SDE dx = -dU/dx * dt + sigma * dW_t near the fold at x_c, the mean first passage time scales as mu_tau ~ (mu_c - mu)^1 / (d_mu/dt) and its standard deviation scales as sigma_tau ~ (mu_c - mu)^(-1/2) * [noise amplitude factor], giving the coefficient of variation CV_tau = sigma_tau / mu_tau ~ (mu_c - mu)^(-3/2) -> infinity. This scaling is derived in Berglund & Gentz (2006, Theorem 5.1) for the canonical fold normal form and confirmed numerically in Dakos et al. (2012). The key references are: Gardiner, C. W. (2004). Handbook of stochastic methods (3rd ed., §5.2). Springer; Berglund, N., & Gentz, B. (2006). Noise-induced phenomena in slow-fast dynamical systems (Ch. 5). Springer; Dakos et al. (2012, PLOS ONE, cited above). The derivation below states the resulting scaling relationship:

sigma_tau(t) proportional to (mu_c - mu(t))^(-1/2)

The expected remaining time to tipping is:

mu_tau(t) proportional to (mu_c - mu(t)) / (d mu/dt) [proportional to distance / forcing rate]

Therefore the coefficient of variation of the tipping time satisfies:

```
CV_tau(t) = sigma_tau(t) / mu_tau(t)
        proportional to (mu_c - mu(t))^(-1/2) / (mu_c - mu(t))
        = (mu_c - mu(t))^(-3/2)
        -> infinity  as  mu(t) -> mu_c  (equivalently, as t -> tau*)
```

THEOREM: CV_tau(t) -> infinity as t -> tau*.

COROLLARY (The Late-Mover Trap):
By Theorem 1, intervention at time t_act is rational iff:

```
E[Delta(t_act)] > C_act * CV_tau(t_act) * mu_tau(t_act) / (C_late - C_act)
```

As t_act -> tau*:

```
LHS: E[Delta(t_act)] = E[tau*] - E[t_signal(t_act)] -> 0  (window narrows)
RHS: C_act * CV_tau(t_act) * mu_tau(t_act) / (C_late - C_act)
    = C_act * (mu_c - mu(t))^(-3/2) * (mu_c - mu(t)) / (C_late - C_act) * [rate factor]
    = C_act * (mu_c - mu(t))^(-1/2) / [(C_late - C_act) * rate_factor]
    -> infinity
```

Since LHS -> 0 and RHS -> infinity, there exists a time t_trap < tau* such that for all t_act > t_trap, the Theorem 1 condition is violated and intervention is irrational.

t_trap is defined implicitly as the unique solution to:

```
E[Delta(t_trap)] = C_act * CV_tau(t_trap) * mu_tau(t_trap) / (C_late - C_act)
```

Proof of existence: The LHS is continuous and strictly decreasing from E[Delta(0)] > 0 to 0.
The RHS is continuous and strictly increasing from a finite value to infinity.
By the Intermediate Value Theorem, a crossing point t_trap exists and is unique
(given that LHS is strictly decreasing and RHS is strictly increasing). [QED]
Note on Theorem 1 form used here: The implicit definition of t_trap above uses the approximate Theorem 1 form (E[Delta] > C_act * CV_tau * mu_tau / (C_late - C_act)). The qualitative IVT existence argument—LHS decreases to 0, RHS increases to infinity, unique crossing exists—holds equally under the rigorous Cantelli form (E[Delta] > CV_tau * mu_tau * sqrt((C_late - C_null) / (C_null - C_act))). The location of t_trap will differ quantitatively between the two forms; practitioners requiring certified estimates should use the Cantelli condition from the Theorem 1 proof above.

INTERPRETATION -- THE CLAUSEWITZ CONNECTION:
Clausewitz observed that friction accumulates as operations approach the culminating point, without identifying the mechanism. Theorem 4 provides the mechanism: the system itself

generates increasing uncertainty about its own transition time as it approaches the bifurcation. This self-generated friction -- not caused by external interference but by the system's own loss of resilience -- creates the Late-Mover Trap. Boyd's response (faster OODA cycling via distributed nodes) is formalized by Theorem 3: distributed detection shifts t_act earlier, equivalent to operating at smaller CV_tau, equivalent to lower self-generated friction.

The paper's central claim is now fully derived: the decision to act must be made before t_trap, the point beyond which self-generated friction makes successful intervention irrational regardless of cost structure.

4.6 Integration of Theorems 1 through 4

The four theorems form a complete decision framework for intervention timing at tipping points. Their relationships are as follows.

Theorem 1 establishes the fundamental condition: intervention is rational iff E[Delta] exceeds a threshold set by the agent's cost structure and the system's transition-time uncertainty. This condition must be satisfied for any intervention to be worth undertaking.

Theorem 2 addresses the agent's single controllable variable: the detection threshold delta_c. Given the cost asymmetry r = C_late/C_fp, the optimal threshold is uniquely determined. A higher r (more asymmetric costs) implies a lower threshold (act on weaker signals). This connects Paper 1's symmetric threshold to the general problem.

Theorem 3 addresses the network architecture. For a given detection technology, distributed monitoring with k independent nodes extends E[Delta] by approximately 1/lambda (one mean detection lead time) in the large-k limit. Since E[Delta] is the crucial quantity in Theorem 1's condition, network architecture directly determines the rationality of intervention.

Theorem 4 addresses the dynamics of the decision problem over time. It proves that the system's own approach to the tipping point degrades the rationality of intervention: CV_tau diverges, pushing the RHS of Theorem 1's condition toward infinity while E[Delta] approaches zero. The intersection of these two trends defines t_trap, the last moment at which intervention is rational.

The Complete Decision Framework (integration of Theorems 1-4)

Given a Transition System S(t), a Sentinel architecture with threshold delta_c, a monitoring network of k independent nodes, and a cost structure (C_fp, C_act, C_late, C_null):

Step 1 (Theorem 2): Set delta_c* = delta_c^(sym) - sigma_delta * ln(r) / sqrt(2*pi)
where r = C_late / C_fp. This is the optimal threshold for the given cost asymmetry.

Step 2 (Theorem 3): Assess the distributed detection advantage.
E[Delta_k] approximately E[Delta_1] + 1/lambda for large k.
If E[Delta_k] is insufficient for Theorem 1's condition, add monitoring nodes.

Step 3 (Theorem 1): Verify E[Delta_k] > C_act * CV_tau * mu_tau / (C_late - C_act).
If the condition is satisfied, intervention is rational at first Sentinel signal.

Step 4 (Theorem 4): Monitor t_trap continuously.
t_trap decreases as the system approaches tau*.
Once t_trap = T_obs (present time), the window has closed.
Any subsequent intervention is irrational under the cost structure.

The FRM Sentinel implements this framework: delta_c from Theorem 2, distributed nodes from Theorem 3, continuous monitoring of the CV_tau trajectory from Theorem 4, and rationality assessment from Theorem 1.

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5. Connection to the FRM Sentinel Architecture

5.1 Sentinel as Virtù Instrument

The FRM Sentinel, introduced in Paper 4 and extended in FRM v5, was designed as a regime-shift detector. This section shows that it is, in formal terms, a virtù instrument: the computational implementation of the decision framework derived in Section 4.

The Sentinel monitors DFA alpha(t) on a rolling window of the substrate's outcome series. When alpha(t) exceeds the threshold delta_c, the Sentinel generates a regime-signal that triggers the KVS-hat adjustment. In the language of Definition 4.3, this is exactly the detection event t_signal. The Sentinel's rolling delta(t) trajectory is the observable proxy for the Fortuna Process F(t), related through Lemma 4.1.

The Sentinel's detection threshold delta_c, calibrated in Paper 4 as a fixed multiple of the stable-regime standard deviation, is now given its proper theoretical grounding by Theorem 2: it is the optimal threshold under the cost asymmetry of the specific substrate. For governance substrates where the cost of missing a regime shift materially exceeds the cost of a false alarm (the typical case), Theorem 2 prescribes a threshold lower than the symmetric default -- i.e., the Sentinel should be more sensitive than a symmetric analysis would suggest.

5.2 FRM Axioms 1 through 4 as Network Structure

FRM Axioms 1 through 4 -- Substrate Independence, Rhythmic Recurrence, Cross-Frequency Coupling, and Fractal Self-Similarity -- specify the measurement and monitoring architecture of the FRM. This section shows that these axioms collectively implement the distributed network structure of Theorem 3.

Axiom 1 (Substrate Independence) ensures that the FRM can monitor multiple substrates simultaneously, each generating an independent Sentinel node. The monitoring of an investment policy substrate, a clinical protocol substrate, and an engineering standards substrate

simultaneously produces k = 3 independent monitoring nodes, with detection window advantage E[Delta_3] approximately E[Delta_1] * 2.

Axiom 2 (Rhythmic Recurrence) specifies that the governance loop cycles at fixed intervals. This is the OODA cycle rate: the frequency at which S(t) is observed and delta(t) updated. Higher cycling frequency increases lambda (the single-node detection rate), directly extending E[Delta_1] and therefore E[Delta_k].

Axiom 3 (Cross-Frequency Coupling) specifies that governance operates simultaneously at multiple timescales: daily operational signals, monthly tactical signals, quarterly strategic signals. This is not three substrates but one substrate monitored at three resolutions -- a form of distributed monitoring in frequency space. Each frequency band provides an independent detection pathway, extending the effective k beyond the number of organizational substrates.

Axiom 4 (Fractal Self-Similarity) specifies that the governance structure at each scale is self-similar: the same loop structure applies at the object level and the meta level. This is the architectural property that makes the network robust to individual node failure -- a property formalized in Paper 2's network stability proof and now grounded in Theorem 3's window extension result.

5.3 Deferred Questions Resolved: The Series Integration

Each of Papers 1 through 4 deferred a question it could not answer. This section states each deferral and its resolution.

Paper and Deferred Question	Resolution in Paper 5
Paper 1: The symmetric loss assumption (kappa = 0.50). What if false positives and false negatives do not cost the same?	Theorem 2 derives the optimal threshold delta_c*(r) for any asymmetry ratio r = C_late/C_fp. Paper 1's kappa = 0.50 is the special case r = 1. The Corrections Register documents this as an intended generalization.
Paper 2: Why are distributed governance networks more resilient? The stability proof showed that they are; it did not explain the mechanism.	Theorem 3 provides the mechanism: k independent monitoring nodes extend the expected virtù window by approximately 1/lambda. Distributed networks are more resilient because they detect transitions earlier. Earlier detection is the source of the advantage.
Paper 3: What triggers the decision to adapt when a signal emerges? The memory architecture was specified; the decision trigger was not.	Theorem 4's t_trap is the decision trigger: the agent must act before the window closes. The memory architecture enables the estimation of t_trap by providing the historical baseline against which CV_tau is assessed.
Paper 4: Once a regime shift is detected by RDS or CSS, when should the agent act? The detection architecture was specified; the response timing was not.	Theorems 1 through 4 together answer this question completely. Act when E[Delta_k] > C_act * CV_tau * mu_tau / (C_late - C_act) (Theorem 1), with detection threshold set by Theorem 2, monitoring network designed by Theorem 3, and t_trap monitored continuously by Theorem 4.

Paper and Deferred Question	Resolution in Paper 5
FRM v5: The EWS literature detects transitions but has no decision theory. What provides the decision theory?	This paper. The FRM Sentinel is a virtù instrument. The four theorems are the decision theory. The series is now complete.

The table above is not merely organizational housekeeping. It demonstrates that Paper 5 was not appended to a complete series; it completes a series that was structurally incomplete without it. The deferred questions are not tangential; they are the questions that every reader of Papers 1 through 4 would eventually ask. Paper 5 is the answer.

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6. Empirical Illustration: AMOC as Pre-Specified Test Case

6.1 Pre-Specification Statement

This section constitutes a pre-registered empirical test of the framework developed in Sections 4 and 5. The test criteria, variables, and success conditions are stated here before any data analysis is conducted. Data results will be reported in a subsequent working paper (Brennan 2026, forthcoming) following completion of the FRM Sentinel data runs specified in Grok Work Order 001.

The Atlantic Meridional Overturning Circulation (AMOC) is selected as the test case for three reasons. First, Boers (2021) provides a published EWS analysis of the RAPID array fingerprint data (2004-2023), with explicit detection of rising variance and autocorrelation consistent with critical slowing down. Second, the AMOC is a fold-bifurcation candidate system with documented hysteresis (Rahmstorf 2002; Caesar et al. 2021), satisfying Definition 4.1. Third, the stakes are high: AMOC collapse would produce substantial climate disruption across the North Atlantic basin, creating a cost asymmetry r >> 1 that makes Theorem 2 directly applicable.

6.2 Formal Mapping to the Framework

S(t) = AMOC fingerprint index (Caesar et al. 2021 sea surface temperature proxy), 1870-2023.

F(t) = Rolling variance of S(t) on a 30-year window (consistent with Boers 2021 methodology).

delta(t) = FRM Sentinel DFA alpha(t) computed on 20-year rolling window of RAPID array data, with alpha_baseline estimated from the 1870-1950 pre-industrialization period.

t_signal = First time delta(t) > delta_c*(r), where r is estimated from the IPCC AR6 cost assessment for AMOC disruption scenarios.

tau* = Unknown. The test asks whether the FRM Sentinel's t_signal is consistent with Boers' (2021) EWS detection timeline.

6.3 Pre-Specified Success Criteria

Test 1 (Cross-validation): The FRM Sentinel's t_signal is within 5 years of the Boers (2021) EWS detection timeline. This constitutes cross-method validation of the pre-tipping classification.

Test 2 (Threshold consistency): The optimal delta_c*(r) computed from Theorem 2 using the IPCC cost estimate for r produces a detection event that matches the directionality of the Boers finding (both flag pre-tipping or both flag stable).

Test 3 (t_trap assessment): The current CV_tau estimated from the RAPID array delta(t) trajectory satisfies the Theorem 4 condition for t_trap < 2030 under the IPCC high-end forcing scenario. This would constitute an actionable claim about intervention timing.

Failure modes: If the FRM Sentinel and Boers (2021) disagree on pre-tipping classification, this is a genuine finding -- either the Sentinel is miscalibrated for oceanographic data or the EWS literature's variance-based metrics overidentify precursors. Either finding is publishable. The test is designed so that both outcomes advance knowledge.

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7. Discussion and Limitations

7.1 The Universality Claim and Its Limits

The five-structure convergence claim is universal in a specific, bounded sense: it applies to systems that (a) exhibit fold-bifurcation tipping behavior, (b) generate observable CSD precursors, and (c) are monitored by agents with the ability to intervene before the tipping point. This covers a broad range of important systems: climate tipping elements, financial stability thresholds, ecological regime shifts, and organizational regime changes. It does not apply to noise-induced tipping (n-tipping), rate-induced tipping (r-tipping), or systems where P(Delta > 0) is substantially below 1. The paper treats these as limiting cases and does not claim the framework applies to them.

The historical convergence claim is similarly bounded. The paper identifies five structural elements that appear in all seven traditions. It does not claim that these traditions' full conceptual architectures are equivalent, or that their specific recommendations translate directly to modern governance contexts. The convergence is structural, not substantive.

7.2 The Three-Scalar Loss Function

The loss function of Definition 4.5 reduces an inherently complex decision problem to three cost parameters. Real systems have multi-attribute loss functions: AMOC collapse affects agriculture, energy, geopolitics, and biodiversity in ways that cannot be reduced to a single C_late. The simplification is deliberate and produces tractable results. The qualitative conclusions of Theorems 1 through 4 are robust to reasonable generalizations of the loss structure, because they depend only on the cost ordering (C_late > C_null > C_act > C_fp), not on specific values. Organizations implementing the framework in practice will need to construct domain-specific loss functions; the theorems characterize the structural conditions rather than prescribing specific parameter values.

7.3 Tipping Cascades

The paper treats a single transition system S(t) with a single tipping time tau*. Armstrong McKay et al. (2022) document the existence of tipping cascades: coupled systems where the transition of one element can trigger transitions in others. Theorem 3's distributed network result extends naturally to multi-system monitoring -- k nodes monitoring m coupled systems produce a joint

detection architecture -- but the full decision theory for cascading tipping points requires treating tau* as a vector rather than a scalar. This extension is left as future work.

7.4 The Empirical Validation Gap

The AMOC test case in Section 6 is pre-specified but not yet executed. The series as a whole rests on mathematical foundations that are more developed than their empirical support. This is an acknowledged limitation that the authors are actively addressing through the data runs specified in Grok Work Order 001. The mathematical framework's value does not depend on the empirical validation; the theorems are true or false on their formal merits. But the claim that the FRM Sentinel provides actionable intervention timing guidance in real systems requires empirical support that this paper initiates but does not complete.

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8. Anticipatory Referee Responses

8.1 On the Historical Sections

Anticipated objection: 'The historical analogies are superficial. Sun Tzu's shi is a metaphor, not a mathematical object.'

Response: The paper makes no claim that Sun Tzu was doing dynamical systems theory. The claim is that the qualitative structure of effective action under transition uncertainty that these traditions independently described is isomorphic to the formal decision-theoretic structure derived from first principles in Section 4. Isomorphism of structure is not identity of intent. The mathematics stands entirely independently of the historical interpretation: remove Sections 3.3 through 3.5 and the theorems remain valid. The historical convergence is corroborating evidence for universality, not the foundation of the mathematical claims. Section 3.1 states this explicitly.

8.2 On the Definition of tau*

Anticipated objection: 'The virtù window Delta = tau* - t_signal is not well-defined because tau* is unobservable.'

Response: Correct, and the paper acknowledges this directly. tau* is treated as a random variable with distribution F_tau estimated from the Sentinel delta trajectory. The virtù coefficient psi is defined in terms of expected costs under this distribution, not in terms of tau* directly. This is standard sequential decision theory under uncertainty (Wald 1947). The unobservability of tau* is not a problem to be solved; it is the defining feature of the problem. Theorem 1's condition is stated in terms of E[Delta], which is estimable from observable quantities. Theorem 4 characterizes how E[Delta] evolves as the system approaches tau*.

8.3 On the Relationship to Wald (1947)

Anticipated objection: 'This is just sequential hypothesis testing with a new name. What is the actual contribution?'

Response: Sequential hypothesis testing provides the mathematical apparatus. The contributions are threefold. First, application to tipping point detection where the alternative hypothesis is non-

stationary: the Type II error probability increases continuously as the system approaches tau*, requiring explicit treatment that standard SPRT does not provide. Second, Theorem 4 (self-generated friction), which derives from CSD scaling the result that CV_tau diverges as t approaches tau*, and proves the existence of t_trap. This result is not in Wald. Third, Theorem 3 (distributed network), which connects the order statistics of k independent exponential detection times to the window extension in governance networks. The connection between network architecture and detection window is not in the decision theory literature.

8.4 On the Dowding System as Centralized

Anticipated objection: 'The Dowding system was centralized, not distributed. All information flowed to Bentley Priory. This contradicts the distributed network claim.'

Response: This objection conflates the sensing architecture with the command architecture. The Dowding system was distributed in its sensing layer (51+ radar stations, 1,000+ Observer Corps posts) and centralized in its orientation layer (Bentley Priory Filter Room). Theorem 3 concerns the sensing layer: k independent sensing nodes reducing the minimum detection time. Park's Rule (Theorem 2 corollary) operates at the command layer. The two layers are explicitly distinct in the paper's treatment. The historical record is unambiguous: the RAF's tempo advantage came from having more independent sensing inputs fused faster -- the log(k) effect. Destroying one radar tower did not blind the system. Destroying the Filter Room would have. This is the architecture Theorem 3 formalizes.

8.5 On Machiavelli's virtù

Anticipated objection: 'The paper misreads Machiavelli. Virtù is not primarily about timing -- it is about character, capacity, excellence.'

Response: The paper does not claim virtù means only timing. It claims that virtù includes the capacity to act effectively within fortune's window -- a reading that is textually supported throughout The Prince and the Discourses, and that is consistent with mainstream Machiavelli scholarship (Mansfield 1996; Skinner 1981). The timing dimension is explicit in Chapter 25, where Machiavelli writes that fortune favors those who act promptly when the opportunity presents, and that hesitation after the opportunity passes is fatal. The paper cites this passage directly and positions it within the broader virtù literature. The mathematical formalization extracts the timing dimension for formal treatment without claiming it is the entirety of virtù.

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9. Conclusion: The Complete Story

9.1 What the Series Built

The Meta-Kaizen series set out to address a specific problem: how should organizations systematically evaluate and adopt improvements to their own governance frameworks? Paper 1 answered: by scoring candidates against a set of axiomatic criteria and approving those above a Bayesian-derived threshold. The answer was clean, mathematically grounded, and self-applicable -- the framework could be applied to itself.

That self-applicability turned out to be the series' most generative property. Applying Meta-Kaizen to its own development produced the bisociation mechanism, which generated Paper 2 (network scaling), Paper 3 (memory architecture), Paper 4 (regime-shift detection), and the climate application of FRM v5. Each bisociation domain contributed something the prior papers lacked. The series grew not by plan but by following the logic of self-improvement to its natural conclusions.

The conclusion it reached is this paper. A governance framework that can detect when its operating environment is changing irreversibly -- which is what the FRM provides -- is incomplete without a decision theory for when to act on that detection. Paper 5 provides that theory.

9.2 What the Mathematics Reveals

The four theorems of Section 4 reveal something that the strategic traditions could see qualitatively but could not make precise: the decision to act is not merely a matter of courage, judgment, or political will. It is a mathematical problem with a determinate structure. The agent who possesses a Sentinel (Theorem 2's delta_c*), deploys it across a distributed network (Theorem 3's k nodes), and monitors the system's self-generated friction (Theorem 4's CV_tau trajectory) has, for the first time, a complete quantitative framework for intervention timing.

The traditions knew this framework existed. Sun Tzu called it shi and jie duan. Machiavelli called it virtù and fortuna. Clausewitz called it the culminating point and friction. Dowding built it in steel, copper, and mathematics he did not have. Boyd formalized its information-theoretic core but not its decision theory. The series provides the decision theory.

9.3 Why the Convergence Matters

Seven traditions, separated by 2,500 years and six civilizations, all found the same five-part structure for acting effectively when a system is undergoing nonlinear transition under uncertainty. This convergence is not a curiosity. It is evidence.

If the five-part structure were arbitrary -- a particular solution to a particular problem -- we would not expect it to appear independently in China, Greece, Florence, Prussia, Britain, and the United States over two and a half millennia. The convergence suggests that the structure is not arbitrary: it is the solution to a problem whose formal structure is universal. Systems approaching critical transitions behave the same way regardless of whether they are armies, republics, ecosystems, or governance frameworks. The agents who act successfully within those transitions follow the same structure regardless of whether they are generals, princes, or portfolio managers.

The mathematics is the explanation for the convergence. The five structural elements correspond to five formal requirements that any rational agent facing a fold-bifurcation tipping point must satisfy. The traditions found these requirements through observation and experience. The series derives them from first principles. The findings are the same because the problem is the same.

9.4 Open Problems

Three open problems are identified for future work. First, the empirical validation of the AMOC test case (Section 6) remains pending. Completion of Grok Work Order 001 and the resulting data analysis will constitute the first full empirical test of the framework.

Second, the extension to tipping cascades (Section 7.3) requires a vector-valued tau* and a multi-agent version of the Virtù Coefficient. The formal architecture is tractable; the application to climate tipping cascade networks (Armstrong McKay et al. 2022) would constitute a significant contribution to the EWS literature.

Third, the behavioral implications of t_trap deserve formal treatment. What institutional mechanisms can extend t_trap -- i.e., preserve the rationality of intervention as systems approach their tipping points? Theorem 3 suggests distributed monitoring. Theorem 2 suggests lower detection thresholds. But the political economy of acting on early signals, before the transition is obvious, remains an open problem that the mathematics of this paper motivates but does not resolve.

These are problems worth having. The series that began by asking how organizations improve their governance frameworks has ended by contributing to the theory of how rational agents act at the boundary of irreversible change. That is a longer journey than was anticipated. It arrived at the right destination.

Corrections Register

The Corrections Register documents deliberate departures from prior papers in the series, significant clarifications of the framework, and resolutions of previously deferred assumptions. It is intended to provide a transparent record for researchers building on this work.

Correction 1: Paper 1's symmetric loss assumption formally extended

Paper 1 derived the KVS approval threshold kappa = 0.50 under a symmetric loss assumption: false positives (approving a low-value candidate) and false negatives (rejecting a high-value candidate) were assumed to carry equal expected costs. This assumption was noted in Paper 1 as a simplification and was not presented as a claim about the real-world cost structure.

Theorem 2 of this paper extends Paper 1's result to the general asymmetric case. The symmetric threshold kappa = 0.50 (equivalently, delta_c^(sym)) is recovered as the special case r = 1 (C_late = C_fp). For r > 1, the optimal threshold is strictly lower than the symmetric threshold.

This is not a correction to Paper 1 in the sense of identifying an error. It is an intended generalization that Paper 1 deferred. Implementing organizations should assess the cost asymmetry of their specific substrate and calibrate delta_c accordingly using Theorem 2.

Correction 2: Paper 2's distributed network advantage now formally grounded

Paper 2 proved that distributed Meta-Kaizen networks are more stable than centralized ones, in the sense that the governance loop closes more reliably across distributed nodes. The proof was formal but the mechanism was not explained: why does distribution provide resilience?

Theorem 3 of this paper provides the mechanism. Distributed monitoring nodes extend the expected virtù window by approximately 1/lambda (one mean single-node detection lead time) in the large-k limit. The resilience advantage is a detection advantage: distributed networks identify approaching regime shifts earlier because the minimum of k independent detection times falls below the single-node detection time.

This grounding does not modify Paper 2's results; it explains them.

Correction 3: Paper 4's detection architecture now connected to intervention timing

Paper 4 specified the FRM's detection architecture (RDS and CSS signals, delta_c threshold) and the adaptive response (KVS-hat, Axiom 5-prime) once a regime signal is received. It did not address the decision timing problem: when, after a regime signal, should the implementing organization act? This deferral was explicit in Paper 4's discussion section.

Theorems 1 through 4 of this paper provide the answer. The complete decision framework of Section 4.6 specifies how to determine whether the current moment satisfies the Theorem 1 rationality condition, how to set delta_c using Theorem 2, how to extend the detection window using Theorem 3, and how to monitor t_trap using Theorem 4.

Paper 4 built the alarm. This paper specifies when to leave the building.

Correction 4: Series completion -- deferred questions resolved

Papers 1 through 4 each deferred one question that the paper could not answer within its scope. These deferrals were not identified as such in those papers; they are named here retrospectively as part of the documentation of the series' intellectual development.

Paper 1 deferred: asymmetric loss. Resolved: Theorem 2.
Paper 2 deferred: mechanism of distributed resilience. Resolved: Theorem 3.
Paper 3 deferred: decision trigger for adaptive action. Resolved: virtù window and t_trap.
Paper 4 deferred: when to act after regime detection. Resolved: Theorems 1 through 4.

The series is now structurally complete. Future papers in the ecosystem (empirical validation, tipping cascade extensions, behavioral implications of t_trap) will extend the framework rather than complete it.

Correction 5: Theorem 4 first-passage time scaling — citation and derivation sketch added

Prior version stated sigma_tau(t) proportional to (mu_c - mu(t))^(-1/2) without citation or derivation. This result follows from first-passage time theory for fold bifurcations under slow parameter forcing, specifically Kramers-type escape rate scaling near the bifurcation point. Theorem 4 now includes a derivation sketch citing Gardiner (2004, §5.2) and Berglund & Gentz (2006, Ch. 5), which provide the canonical treatment of this scaling. Both references added to the reference list. The result itself is unchanged; only the supporting citation and derivation sketch are new.

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